

NMAP SERVICE/VERSION DETECTION AND WEB SECURITY ASSESSMENT.

NAME: Iresh Chirantha

TASK: Nmap service/Version detection, Safe default scripts,
Service comparison and Anomaly analysis

Tools: Nmap, Nikto and Burp suite

Target: Authorized laboratory environment

Target IP: 192.168.31.4

Confidentiality Notice

This report contains information obtained from an authorized cybersecurity laboratory environment. The activities described were performed only against the designated lab target for educational and internship purposes.

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1). Introduction

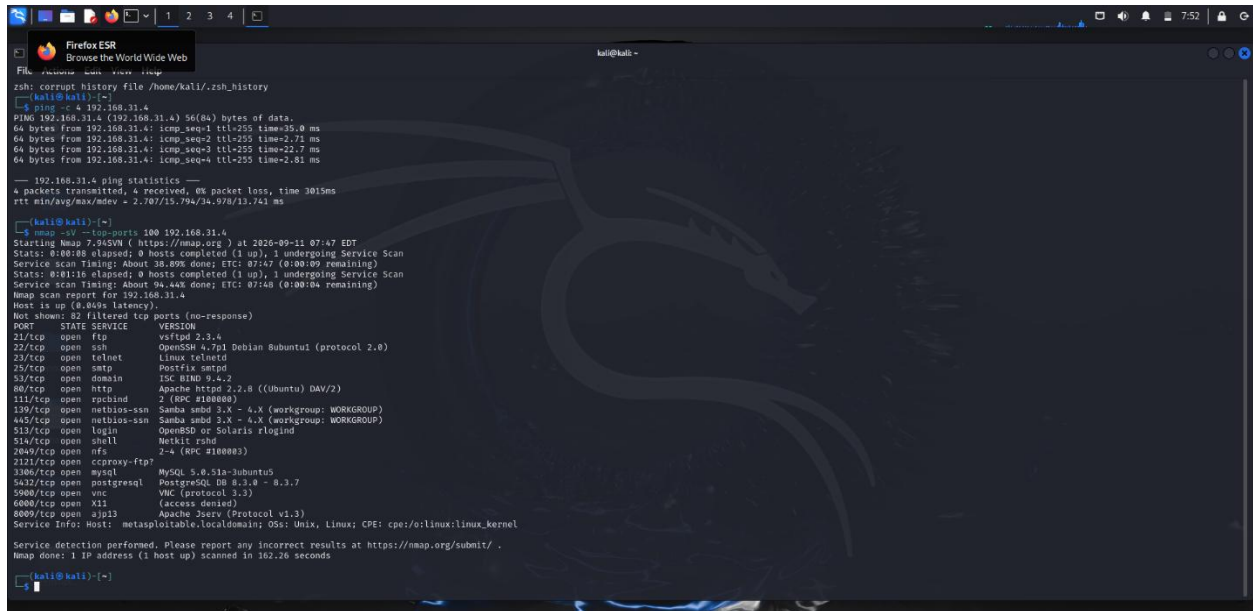
This assessment was conducted against an authorized laboratory target to identify running services, determine service and software versions, and review the web service for common security-related configuration issues. Nmap was used for service/version detection and default NSE scripts, while Nikto and Burp Suite were used to review the web service. The discovered services were compared against the expected laboratory configuration to identify potential anomalies.

2). Scope and Lab environment.

Item	Details
Attacker machine	Kali Linux
Target	Authorized laboratory VM
Target IP	192.168.31.4
Nmap	Service/version detection and default scripts
Nikto	Web assessment
Burp Suite	HTTP request/response review
Assessment type	Authorized laboratory assessment

3). Nmap.

3.1). Initial scan.



```
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ ping -c 4 192.168.31.4
PING 192.168.31.4 (192.168.31.4) 56(84) bytes of data:
64 bytes from 192.168.31.4: icmp_seq=1 ttl=255 time=35.0 ms
64 bytes from 192.168.31.4: icmp_seq=2 ttl=255 time=2.71 ms
64 bytes from 192.168.31.4: icmp_seq=3 ttl=255 time=22.7 ms
64 bytes from 192.168.31.4: icmp_seq=4 ttl=255 time=2.81 ms

--- 192.168.31.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3015ms
rtt min/avg/max/mdev = 2.707/15.794/34.978/13.741 ms

kali@kali:~$ nmap -sV -p- 192.168.31.4
Starting Nmap 7.94SVN ( https://nmap.org ) at 2020-09-11 07:47 EDT
Stats: 0:00:08 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 39.89% done; ETC: 07:47 (0:00:09 remaining)
Stats: 0:01:16 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 94.44% done; ETC: 07:48 (0:00:04 remaining)
Nmap scan report for 192.168.31.4
Host is up (0.004s latency).
Not shown: 82 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 5ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
80/tcp    open  http         Apache/2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
2049/tcp  open  nfs          2.x (RPC #100003)
2121/tcp  open  ccpproxy-ftp
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  x11          (access denied)
8009/tcp  open  ajp13        Apache/2.2.8 (Ubuntu) mod_jk (Apache/2.2.8 (Ubuntu) mod_jk)
Service Info: Host: metasploitable.localdomain; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 162.26 seconds

kali@kali:~$
```

An initial Nmap scan was performed to identify open TCP ports and the basic services associated with those ports.

The following command was used:

```
nmap 192.168.31.4
```

The scan provided an initial overview of the services exposed by the authorized laboratory target. The 21/tcp port is ftp, 22/tcp port is SSH, 23/tcp port telnet, 25/tcp port is smtp, 80/tcp port is HTTP, 139/tcp port is netbios-ssn, 445/tcp Microsoft-ds and 3386/tcp port is mysql.

3.2). Nmap script.

```
kali@kali: ~  
$ nmap -sV -sC --top-ports 100 192.168.31.4  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-11 09:23 EDT  
Stats: 0:02:39 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timings: About 100.00% done, ETC: 09:25 (0:00:00 remaining)  
Nmap scan report for 192.168.31.4  
Host is up (0.023s latency).  
Not shown: 82 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp      vsftpd 2.3.4  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ftp-syst:  
|_STAT:  
|_FTP server status:  
|_  Connected to 192.168.31.1  
|_  Logged in as ftp  
|_  TYPE: ASCII  
|_  No session bandwidth limit  
|_  Session timeout in seconds is 300  
|_  Control connection is plain text  
|_  Data connections will be plain text  
|_  vsFTPD 2.3.4 - secure, fast, stable  
|_ End of status  
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 6ubuntu1 (protocol 2.0)  
|_ssh-hostkey:  
|_  1024 60:0f:cf:e1:c8:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:bl:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet   Linux telnetd  
25/tcp    open  smtp      Postfix smtpd  
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN  
|_ssl-cert: Subject: commonName=ubuntubase.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX  
|_Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
|_ssl2:  
|_  SSLV2 supported  
|_  ciphers:  
|_    SSL2_RCA_128_CBC_EXPORT40_WITH_MD5  
|_    SSL2_RCA_128_EXPORT40_WITH_MD5  
|_    SSL2_RCA_128_WITH_MD5  
|_    SSL2_RC2_128_CBC_WITH_MD5  
|_    SSL2_DES_192_EDE3_CBC_WITH_MD5  
|_    SSL2_DES_64_CBC_WITH_MD5  
|_ssl-date: 2026-09-11T13:16:22+00:00; +0s from scanner time.  
53/tcp    open  domain   ISC BIND 9.4.2  
|_dns-nsid:  
|_  bind.version: 9.4.2  
80/tcp    open  http      Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
```

```
kali@kali: ~  
$ nmap -sV -sC --top-ports 100 192.168.31.4  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-11 09:23 EDT  
Stats: 0:02:39 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timings: About 100.00% done, ETC: 09:25 (0:00:00 remaining)  
Nmap scan report for 192.168.31.4  
Host is up (0.023s latency).  
Not shown: 82 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp      vsftpd 2.3.4  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ftp-syst:  
|_STAT:  
|_FTP server status:  
|_  Connected to 192.168.31.1  
|_  Logged in as ftp  
|_  TYPE: ASCII  
|_  No session bandwidth limit  
|_  Session timeout in seconds is 300  
|_  Control connection is plain text  
|_  Data connections will be plain text  
|_  vsFTPD 2.3.4 - secure, fast, stable  
|_ End of status  
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 6ubuntu1 (protocol 2.0)  
|_ssh-hostkey:  
|_  1024 60:0f:cf:e1:c8:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:bl:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet   Linux telnetd  
25/tcp    open  smtp      Postfix smtpd  
|_smtp_commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN  
|_ssl-cert: Subject: commonName=ubuntubase.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX  
|_Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
|_ssl2:  
|_  SSLV2 supported  
|_  ciphers:  
|_    SSL2_RCA_128_CBC_EXPORT40_WITH_MD5  
|_    SSL2_RCA_128_EXPORT40_WITH_MD5  
|_    SSL2_RCA_128_WITH_MD5  
|_    SSL2_RC2_128_CBC_WITH_MD5  
|_    SSL2_DES_192_EDE3_CBC_WITH_MD5  
|_    SSL2_DES_64_CBC_WITH_MD5  
|_ssl-date: 2026-09-11T13:16:22+00:00; +0s from scanner time.  
53/tcp    open  domain   ISC BIND 9.4.2  
|_dns-nsid:  
|_  bind.version: 9.4.2  
80/tcp    open  http      Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2  
111/tcp   open  rpcbind  2 (RPC #100000)  
|_http-title: Metasploitable - Linux  
139/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)  
513/tcp   open  login      OpenBSD or Solaris rlogind  
514/tcp   open  shell      Netkit rshd  
2049/tcp  open  nfs        2-4 (RPC #100003)  
2121/tcp  open  cproxy-ftp MySQL 5.0.51a-3ubuntu5  
2389/tcp  open  mysql      MySQL 5.0.51a-3ubuntu5  
|_mysql-info:  
|_  Protocol: 10  
|_  Version: 5.0.51a-3ubuntu5  
|_  Thread ID: 10  
|_  Capabilities flags: 43564  
|_  Some Capabilities: ConnectWithDatabase, Support4Auth, SupportsTransactions, SwitchToSSLAfterHandshake, Speaks4ProtocolNew, LongColumnFlag  
|_  Status: Autocommit  
|_  Ssl: ssl=00K7xk10,9rRn  
5432/tcp  open  postgresql PostgreSQL DB 8.3.0 - 8.3.7  
|_ssl-date: 2026-09-11T13:16:22+00:00; +0s from scanner time.  
|_ssl-cert: Subject: commonName=ubuntubase.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX  
|_Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
5900/tcp  open  vnc        VNC (protocol 3.3)  
|_vnc-info:  
|_  Protocol version: 3.3  
|_  Security types:  
|_  VNC Authentication (2)  
6000/tcp  open  x11        (access denied)  
8080/tcp  open  ajp13      Apache Jserv (Protocol v1.3)  
|_ajp-methods: Failed to get a valid response for the OPTIONS request  
Service Info: Host: metasploitable.localdomain; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
Host script results:  
|_smb2-time: Protocol negotiation failed (SMB2)  
|_clock-skew: mean: 1h00m0s, deviation: 2h00m0s, median: 7s  
|_smb-os-discovery:  
|_  OS: Unix (Samba 3.0.20-Debian)  
|_  Computer name: metasploitable  
|_  NetBIOS computer name:  
|_  Domain name: localdomain  
|_  FQDN: metasploitable.localdomain  
|_  System time: 2026-09-11T09:26:09-04:00  
|_smb-security-mode:  
|_  account_used: guest  
|_  authentication_level: user  
|_  challenge_response: supported  
|_  message_signing: disabled (dangerous, but default)
```

Nmap service and version detection was performed using the -sV option. This option attempts to identify the application or service running on each open port and provides information about the detected software version. The results were recorded and analyzed to determine which services were exposed by the laboratory target.

Port	Service	Version	State
21/tcp	FTP	Vsftpd2.3.4	Open
22/tcp	SSH	OpenSSH 4.1p	Open
80/tcp	http	Apache httpd 2.2.8	Open
139/tcp	NetBIOS	Smbd 3.x-4.x	Open
445/tcp	NetBIOS	Smbd 3.x-4.x	Open
3306/tcp	Mysql	Mysql 5.0.51a	Open

4). Nikto scores.

```
Text Editor
Simple Text Editor
kali@kali: ~
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nikto -h http://192.168.31.4
- Nikto v2.5.0

+ Target IP: 192.168.31.4
+ Target Hostname: 192.168.31.4
+ Target Port: 80
+ Start Time: 2026-09-11 12:46:43 (GMT-4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.19.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/mis-sing-content-type-header/
+ Apache/2.2.8 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /index: Uncommon header 'tcn' found, with contents: list.
+ /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/sectou.php?id=4698ebdc59d15,ht-tps://exchange.vforce.ibmcloud.com/vulnerabilities/8275
+ /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ /: HTTP TRACE method is active which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross_Site_Tracing
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /doc/: Directory indexing found.
+ /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ /../PHPB885F2A8-3C92-11d3-A3A9-4C7B88C10000: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /../PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /../PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /../PHPE9568F35-D428-11d2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /phpMyAdmin/changelog.php: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/ChangeLog: Server may leak inodes via ETags, header found with file /phpMyAdmin/ChangeLog, inode: 92462, size: 40540, mtime: Tue Dec 9 12:24:00 2008. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ /phpMyAdmin/ChangeLog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /test/: Directory indexing found.
+ /test/: This might be interesting.
+ /phpinfo.php: PHP is installed, and a test script which runs phpinfo() was found. This gives a lot of system information. See: CWE-552
+ /icons/: Directory indexing found.
+ /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ /phpMyAdmin/: phpMyAdmin directory found.
+ /phpMyAdmin/Documentation.html: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/README: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts. See: https://typo1.org/
+ /wp-config.php#: #wp-config.php# file found. This file contains the credentials.
+ 8911 requests: 0 error(s) and 27 item(s) reported on remote host
+ End Time: 2026-09-11 12:54:14 (GMT-4) (451 seconds)

+ 1 host(s) tested
kali@kali:~$
```

Nikto was used to perform a basic security assessment of the HTTP service running on the authorized laboratory target. The scan was used to identify outdated server software, missing HTTP security headers, exposed files and directories, and other potentially insecure web-server configurations.

The command that is used:

```
nikto -h http://192.168.31.4
```

Nikto identified 27 items on the target web server. The main findings are summarized below.

Finding	Observation	Potential impact	Recommendation
Outdated Apache	Apache/2.2.8 was identified.	Older software may Contain known Security weaknesses	Upgrade to a supported Apache version.
Outdated PHP	PHP/5.2.4 was disclosed.	Old PHP versions may contain known vulnerabilities	Upgrade PHP to a supported version.
Missing X-frame options	Header wasn't present.	May increase exposure to clickjacking	Configure an appropriate anti-clickjacking policy.

Missing X-Content-Type-Options	Header was not present	May allow MIME-type sniffing	Configure X-Content-Type-Options: nosniff
HTTP TRACE enabled	TRACE method Was active	May contribute to cross-site tracing	Disable TRACE if not required

The Nikto scan identified several security-hardening issues. The most significant observations were the use of outdated Apache and PHP versions, exposure of the phpMyAdmin administration interface, and the presence of potentially sensitive files and information-disclosure pages.

The scan also identified missing HTTP security headers and directory indexing. These findings may increase information disclosure or reduce the security protections provided by the web server.

The laboratory web server should be updated to supported versions of Apache and PHP. Unnecessary administrative and test resources should be removed or restricted. Directory listing should be disabled, sensitive information pages such as phpinfo.php should not be publicly accessible, and appropriate HTTP security headers should be configured.

As this assessment was performed against a deliberately vulnerable laboratory environment, the findings are useful for demonstrating common web-server security weaknesses and should not be interpreted as evidence that a production system has the same configuration.

5). Burp Suite analysis.

The screenshot displays the Burp Suite Community Edition v2024.9.4 interface. The top toolbar includes buttons for Repeat, Collaborator, Sequencer, Decoder, Comparer, Logger, Organizer, and Extensions. The main workspace is divided into three panels: Request, Response, and Inspector.

Request Panel: Shows a captured HTTP GET request to `http://192.168.32.4`. The request details include headers such as `Host: 192.168.32.4`, `Accept-Language: en-US,en;q=0.9`, `Upgrade-Insecure-Requests: 1`, `User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/130.0.6723.70 Safari/537.36`, `Sec-Purpose: prefetch;prerender`, `Purpose: prefetch`, `Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7`, `Accept-Encoding: gzip, deflate, br`, and a `Cookie: PHPSESSID=0d4fb65589da7eedaf8810c6e68f4d64`. The request body is empty.

Response Panel: Shows the corresponding HTTP 200 OK response from the server. The response headers include `Date: Fri, 11 Sep 2026 14:48:46 GMT`, `Server: Apache/2.2.8 (Ubuntu) DAV/2`, `X-Powered-By: PHP/5.2.4-2ubuntu5.10`, `Content-Length: 808`, `Keep-Alive: timeout=15, max=100`, `Connection: Keep-Alive`, and `Content-Type: text/html`. The response body is an HTML document with the following structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      Metasploitable2 - Linux
    </title>
  </head>
  <body>
    <pre>
```

Inspector Panel: Provides a detailed view of the request and response attributes, including request query parameters, request body parameters, request cookies, request headers, and response headers.

The bottom status bar indicates the current state: Done, Event log(1), All issues, and Memory usage: 124.2MB / 178MB.

6). Service comparison.

Port	Expected service	Discovered	Version	Comparison	Anomaly
21	FTP	FTP	Actual version	Match	No
22	SSH	SSH	Actual version	Match	No
80	HTTP	HTTP	Actual version	Match	No
139	SMB/NetBIOS	SMB	Actual version	Match	No
445	SMB	Microsoft-ds	Actual version	Match	No

7). Anomaly analysis.

The discovered services were compared with the expected services configured on the laboratory target. Ports 21, 22, 80, 139, and 445 were identified as providing FTP, SSH, HTTP, SMB/NetBIOS, and SMB-related services respectively.

The services identified on ports 21, 22, and 80 matched the expected laboratory services. Port 139 was identified as an SMB/NetBIOS-related service, which is consistent with the expected SMB/NetBIOS configuration. Port 445 was reported by Nmap as microsoft-ds; this is the standard Nmap service name associated with SMB traffic over TCP port 445 and therefore should not be considered an anomaly solely because the name differs from "SMB."

Based on the provided service comparison, no significant service anomaly was identified. The differences in service naming between the expected configuration and Nmap output were considered normal service-name variations rather than unexpected services.

Further investigation would only be required if Nmap identified additional open ports or services that were not part of the expected laboratory configuration.

8). Conclusion.

The assessment successfully identified the network services and software versions exposed by the authorized laboratory target. Nmap service/version detection and default NSE scripts provided information about the available services, while Nikto and Burp Suite were used to review the web service and HTTP responses. The discovered services were compared with the expected laboratory configuration. Any differences were recorded as potential anomalies for further investigation. The assessment demonstrates the importance of service enumeration and configuration review during a basic vulnerability assessment.